



**Verified Carbon
Standard**

TUMED LEFT BANNER IMPROVED AGRICULTURAL LAND MANAGEMENT PROJECT



Document Prepared by Shanghai Discovery Energy Services Co., Ltd.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Tumed Left Banner Improved Agricultural Land Management Project (hereinafter referred to as “the project”) is located in Tumed Left Banner, Hohhot City, Inner Mongolia Autonomous Region, China. The purpose of the project is to adopt improved agricultural land management practices to prevent the severe soil degradation and salinization caused by pre-existing conventional tillage in local crop land.

In the context of the Project, improved practices are categorized according to practices leading to a change in pre-project agricultural management practices, and these could include: improved fertilizer (increased organic or reduced synthetic) application; soil conditioners application; improved water management/irrigation; improved residue management.

The project plans to implement improved agricultural land management practices in 37,272.2 ha field in the first crediting period to increase soil organic carbon (SOC). The project proponent not only introduced more efficient and appropriate tillage method with Tumed Left Banner Agriculture and Animal Husbandry Bureau (TAAHB) to the farmers, but also provide technical training and support to the farmers.

The scenario existing prior to the project’s implementation is primarily conventional farming and conventional irrigation methods, with less application of organic fertilizers and straw mulching, which increase soil disturbance and the loss of soil organic matter (SOM). The baseline scenario is the same as the conditions existing prior to the project initiation.

During the first crediting period from 22-Sep-2022 to 21-Sep-2042, the project is expected to generate GHG emission removals of 1,530,799 tCO₂e, with an average annual GHG emission removal of 76,540 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	To be listed	VCS	To be listed	NA

1.3 Sectoral Scope and Project Type

Sectoral scope	14. Agriculture, forestry and other land use (AFOLU)
AFOLU project category ¹	Agricultural Land Management (ALM)

¹ See Appendix 1 of the VCS Standard

Project activity type

Improved Cropland Management (ICM)

1.4 Project Eligibility

1.4.1 General eligibility

The Project adopts measures to increase removals of CO₂, a Kyoto Protocol greenhouse gas. The applied methodology VM0042 has been approved under the VCS Program through the methodology approval process. Scope 1) and Scope 3) included in Section 2.1.1 of VCS Standard version 4.7 are hence met, and the other three scopes are irrelevant.

The project activity is not any of the activities excluded in Table 1 of VCS Standard version 4.7.

As per Section 3.8.2 of VCS Standard version 4.7, AFOLU projects shall initiate the pipeline listing process within three years of the project start date. As the Project starts on 22-Sep-2022, the pipeline listing deadline would be 21-Sep-2025. The Project initiates the pipeline listing process before the deadline.

In addition, the project has not generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Therefore, the Project meets the general eligibility requirements.

1.4.2 AFOLU project eligibility

As per Section A1.2 of VCS Standard version 4.7, eligible ALM activities are those that reduce net GHG emissions on croplands and grasslands by increasing carbon stocks in soils and woody biomass and/or decreasing CO₂, N₂O and/or CH₄ emissions from soils. The project area shall not be cleared of native ecosystems within the 10-year period prior to the project start date. The category of Improved Cropland Management (ICM) includes practices that demonstrably reduce net GHG emissions of cropland systems by increasing soil carbon stocks, reducing soil N₂O emissions, and/or reducing CH₄ emissions.

The Project is expected to reduce net GHG emissions on croplands by increasing carbon stocks in soils. The project area belongs to the native ecosystem of cropland. No clearing or conversion took place within at least 10 years of the project start date and the project area has not been converted, cleared, drained or degraded to generate GHG credits.

Therefore, the Project meets the AFOLU-specific eligibility requirements.

1.4.3 Transfer project eligibility

This section is irrelevant as the project is not a transfer project.

1.5 Project Design

This project is a single project activity. The project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

Not applicable as the Project is not a grouped project.

1.6 Project Proponent

Organization name	Tumed Left Banner Aletan Agricultural and Animal Husbandry Development Investment Co., LTD
Contact person	Wang Jianfeng
Title	Chairman
Address	West of Quansheng Road, Chasuqi Town, Tumed Left Banner, Hohhot, Inner Mongolia Autonomous Region, China
Telephone	+86 18686089126
Email	1208737893@qq.com

1.7 Other Entities Involved in the Project

Organization name	Shanghai Discovery Energy Services Co., Ltd.
Role in the project	VCS Consultant
Contact person	Guan Wenyu
Title	Project Manager
Address	10F, Block C, 515 Shennan Road, Minhang District, Shanghai, China
Telephone	+86 021 3468 8219
Email	carbonprojects@discoveryesco.com

1.8 Ownership

According to Article 10 of the Constitution of the People's Republic of China² and the provisions of the Land Management Law of People's Republic of China³, the land belongs to the state or collectives and the ownership is non-transferable, while the land use right can be transferred by law.

The fields in the project area are owned by village collectives (project ownership arising under law) and used by farmer households through family-based contracting. Since these fields are all legal agricultural fields, and the ownership is clear, there is no dispute over land ownership in the project site.

The village collective who owns the project land all agreed to authorize Tumed Left Banner Aletan Agricultural and Animal Husbandry Development Investment Co., LTD (hereinafter "the project proponent") the agricultural land management rights and carbon sink revenue rights of the project area in the whole crediting period, while the project proponent should be responsible for the maintenance and overall management of the improved agricultural land during the whole crediting period. In addition, the project proponent will also be responsible for development of the carbon credits and ensure that the carbon revenue will be used for sustainable development and management of the project activities.

Shanghai Discovery Energy Services Co., Ltd. as a consultancy is responsible for helping project proponent in the development and management of carbon projects.

1.9 Project Start Date

Project start date	22-Sep-2022
Justification	According to VCS standard, v4.7, the project start date is the date on which activities that lead to the generation of GHG emission reductions or removals are implemented. The start date of improved agricultural land management is 22-Sep-2022. Thus, the start date of the project is 22-Sep-2022.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable
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² http://www.gov.cn/guoging/2018-03/22/content_5276318.htm

³ http://www.npc.gov.cn/npc/c2/c30834/201909/t20190905_300663.html

	☐ <i>Ten years, fixed</i> ☒ <i>Other</i> According to paragraph 3.9.2 and 3.9.3 of VCS standard version 4.7, for ALM projects not focusing exclusively on reducing N₂O, CH₄ and/or fossil-derived CO₂ emissions, the project crediting period shall be a minimum of 20 years up to a maximum of 100 years, which may be renewed at most four times with a total project crediting period not to exceed 100 years. Moreover, according to paragraph 3.2.5 of VCS Standard version 4.7, for all ALM projects, the project proponent shall, for the duration of the project, reassess the baseline every ten years. Thus, the project adopts 20 years, three times renewable crediting period for a total of 80 years. The first crediting period of the project is 20 years from 22-Sep-2022 to 21-Sep-2042.
	Start and end date of first or fixed crediting period
	22-Sep-2022 to 21-Sep-2042

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
22-Sep-2022 to 31-Dec-2022	8,948
01-Jan-2023 to 31-Dec-2023	38,205
01-Jan-2024 to 31-Dec-2024	56,266
01-Jan-2025 to 31-Dec-2025	80,537
01-Jan-2026 to 31-Dec-2026	80,537
01-Jan-2027 to 31-Dec-2027	80,537
01-Jan-2028 to 31-Dec-2028	80,537

01-Jan-2029 to 31-Dec-2029	80,537
01-Jan-2030 to 31-Dec-2030	80,537
01-Jan-2031 to 31-Dec-2031	80,537
01-Jan-2032 to 31-Dec-2032	80,537
01-Jan-2033 to 31-Dec-2033	80,537
01-Jan-2034 to 31-Dec-2034	80,537
01-Jan-2035 to 31-Dec-2035	80,537
01-Jan-2036 to 31-Dec-2036	80,537
01-Jan-2037 to 31-Dec-2037	80,537
01-Jan-2038 to 31-Dec-2038	80,537
01-Jan-2039 to 31-Dec-2039	80,537
01-Jan-2040 to 31-Dec-2040	80,537
01-Jan-2041 to 31-Dec-2041	80,537
01-Jan-2042 to 21-Sep-2042	58,251
Total estimated ERRs during the first or fixed crediting period	1,530,799
Total number of years	20
Average annual ERRs	76,540

1.12 Description of the Project Activity

There is no REDD+ component to the project and the project is not located within a jurisdiction covered by any jurisdictional REDD+ program.

The proposed ALM project activities aim to promote a variety of environmental and climate-friendly agricultural practices, mainly including soil improvement and irrigation, to enhance soil carbon sinks for agricultural management and saline-alkali land remediation.

The project plans to implement the improved agricultural land management practices in 37,272.2 ha field in Tumed Left Banner in the first crediting period.

Before the project was implemented, the project initiator cooperated with the TAAHB to provide farmers with corresponding agricultural technology training, including the application methods of soil organic fertilizers and soil conditioners, as well as the key points of water-saving irrigation measures management, to ensure the effective implementation of the project.

During the project implementation period, the project owner cooperated with the TAAHB to carry out the construction of the drainage and irrigation pipeline system and drip irrigation measures, complete the selection and application of organic fertilizers and soil conditioners , and arrange the straw returning to the field after the harvest. The project owner not only provides corresponding technical training for farmers, but also supervises them in soil improvement, irrigation and crop residue management operations, and checks the completion of various measures. Farmers are the main operators who practice these improved agricultural land management measures and are responsible for recording their cultivation practice in logbooks provided by the project proponent. The technicians should carry out periodic inspections according to the requirements in Monitoring and Quality Control Manual.

The project start date is 22-Sep-2022. The project activity is expected to generate GHG emission removals of 76,540 tCO₂e per year and contributes to 3 SDGs. Please refer to section 1.18 for the details.

The table below shows the Practice Categories implemented on the initial project activity instances, during the first monitoring period. The table shows the changes adopted, corresponding conventional farming practices, and key-characteristics and measures employed that would lead to reduction or removal of GHG emissions.

Practice Category	IALM Changes (>5%)	Baseline activities (conventional agriculture)	Key- Characteristics & measures employed
Improved fertilizer use (organic or inorganic) application	Organic fertilizer use	Before the implementation of the project, the application rate of organic fertilizer was low, and there were no scientific standards for the sources and application methods of the fertilizer.	Well-rotted farmyard organic fertilizer contains a large amount of organic carbon. After being applied to the soil, it can directly replenish the carbon source for the soil. Continuous application can form a virtuous cycle of "soil-plant": After the organic carbon in the soil increases, crops grow more vigorously, absorb more CO ₂ through photosynthesis, and when the crop residues return to the soil, they can further replenish the carbon source.
Soil conditioners application	Organic improvers: humic acid	Traditional farming methods do not apply soil conditioners.	Organic soil conditioners play a role in transforming saline-alkali soil. After humic acid and microorganisms are applied to the

			soil, they can directly increase the organic carbon content in the soil, promote the formation of soil aggregate structure, enhance carbon sequestration capacity.
Improved water management/irrigation	Build irrigation and drainage systems and carry out earthen canal lining. Introduce water-saving and precision agriculture technologies, such as sub-film drip irrigation and water and fertilizer integration technology.	Before the project was implemented, it mainly relied on flooding irrigation, which not only wasted water resources but also led to the destruction of soil structure and the loss of nutrients. The various drainage and irrigation facilities are not complete, and saline-alkali land has formed in many areas within the banner.	By means of dual irrigation through Wells and channels and the construction of field drainage facilities, the salt-rich layer is controlled below the root layer of crops, the physical and chemical structure of the soil is improved, and the accumulation of soil organic carbon is promoted. Water-saving measures such as drip irrigation can save a large amount of water resources and the consumption of water pumping energy (such as diesel pumps and electricity).
Residue Management	Crop residue left on the land to decay.	The implementation rate of returning crop residue on the land is low, and some crop residue is directly burned.	Leaving agricultural residue in field after harvesting and returning plant material to the soil.

1.13 Project Location

This project is located in Tumud Left Banner, Hohhot City, Inner Mongolia. It includes Shaerqin Town, Beishizhou Township, Shandai Town, Tabusai Township, Baimiaozi Town, Bikeqi Town, Chilechuan Town, and Chasuqi Town. The geographical coordinate range is: 111° 42' to 111° 49' east longitude and 40° 36' to 40° 41' north latitude.

The location of the project activity in Tumud Left Banner which is illustrated in Figure 1-1. And the KML file has also been submitted.

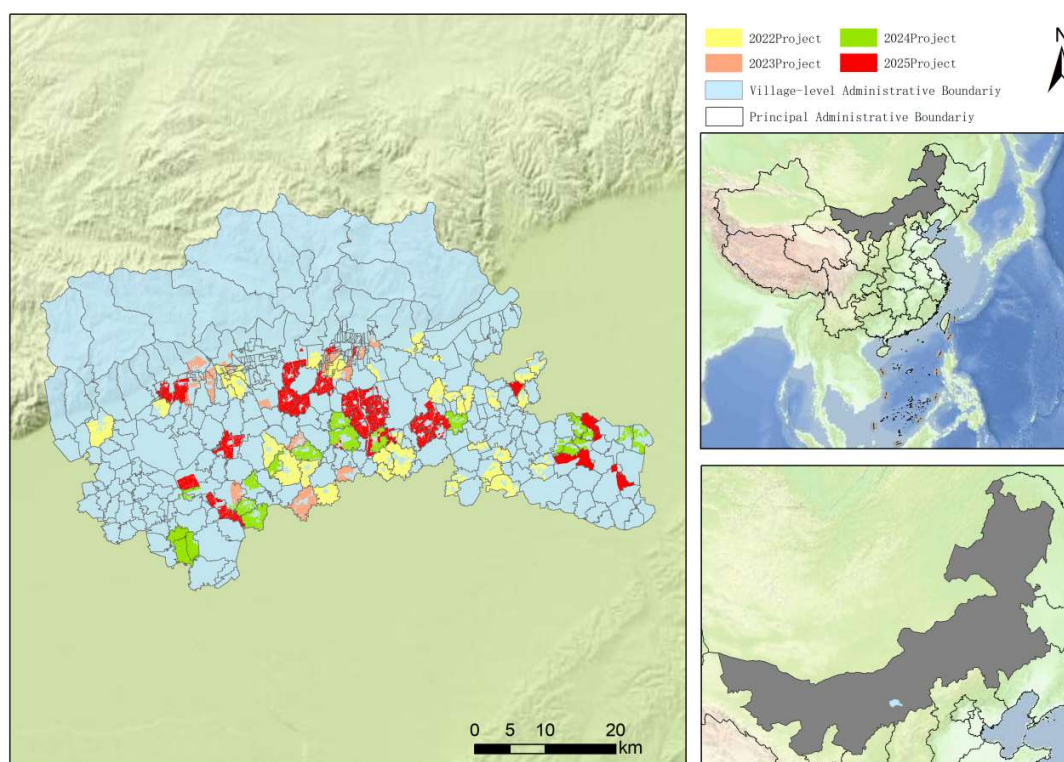


Figure 1-1

1.14 Conditions Prior to Project Initiation

The conditions existing before the project initiation are the same as the baseline scenario, please refer to Section 3.4 (Baseline Scenario).

Ecosystem type: The project area exclusively belongs to the cropland ecosystem.

Current and historical land-use: Supervised classification using historic satellite images of the region has been applied to show that the project area has been cropland for at least 10 years prior to the project start, i.e., from 2012 to 2022.

The following table shows the present and prior environmental conditions of the project area:

Condition	Description
Climate	Tumed Left Banner has a mid-temperate semi-arid continental monsoon climate with distinct four seasons. According to Volume 4 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories It belongs to the Cool temperate moist climate zone. Spring is

	dry and windy, summer is hot and less rainy, autumn is cool and short, and winter is cold and dry. The annual average temperature is about 6.3 °C, and the annual average precipitation is 330-410 mm, with most precipitation concentrated in July-September. The annual average sunshine hours are 2,876, and the frost-free period is 130-140 days. The temperature difference between day and night is relatively large, and the climatic conditions have a certain impact on agricultural production, which requires relying on irrigation to ensure crop growth⁴.
Hydrology	The main rivers in Tumed Left Banner include the Dahei River, Xiaohei River, and Shila Wusu River, all belonging to the Yellow River system. These rivers provide an important water source for local agricultural irrigation, industrial water use, and domestic water use. In addition, there are some lakes and reservoirs, such as Hasuhai. The groundwater resources in the region are relatively abundant, but the distribution is uneven, and there is shallow groundwater salinization in some areas.
Topography	Tumed Left Banner is located in the southern part of the Inner Mongolia Plateau, with the terrain mainly being plain and a small amount of mountains and hills in the north. The overall terrain is high in the north and low in the south, with an average altitude of about 1,000 meters. The plain area is flat and open, being the main agricultural production area; the northern mountainous and hilly areas have relatively small terrain undulations and relatively little vegetation coverage.
Soils	The main soil types in Tumed Left Banner include chestnut soil, chao soil, saline soil, and aeolian sandy soil. Chestnut soil is widely distributed and is one of the main agricultural soils, but it has low organic matter content and relatively weak fertility. Chao soil is mainly distributed along rivers, with good soil moisture conditions, but some areas have salinization problems. Saline soil and aeolian sandy soil are distributed in local areas, with low soil productivity and requiring improvement before they can be used for agricultural production.

⁴ *Design documents for improved agricultural land management in Tumed Left Banner*

Relevant historical conditions	Tumed Left Banner has long been dominated by nomadic herding. During the Qin and Han dynasties, a small number of people from the Central Plains moved in, bringing about small-scale farming. During the reign of Altan Khan in the Ming Dynasty, Han people were encouraged to emigrate and build water conservancy projects, and some grasslands gradually turned into farmland. The Qing Dynasty's "westward Migration" led to a large influx of Han people. During the Qianlong period, the arable land in the Tumed Plain was basically reclaimed, and the economy shifted towards a balance between agriculture and animal husbandry. In modern times, especially after the founding of the People's Republic of China, agricultural infrastructure has been continuously improved and mechanization has been advanced. The scale and output of farmland have significantly increased. Today, it has become one of the important grain-producing areas in Inner Mongolia.
Type of vegetation	The vegetation within the project area is dominated by agricultural vegetation, which primarily consists of crops like corn and sunflowers—these are widely distributed across the plain agricultural regions. Besides, a small number of arbor and shrub species can be found along rivers and around villages, with examples including poplar, willow, and sea buckthorn.
Ecosystems	The project area exclusively belongs to the cropland ecosystem, which is centered around crops such as corn and sunflowers. Relying on the local cool temperate climate and irrigation conditions, a stable planting pattern has been formed. In addition to contiguous crop communities, the system is also equipped with infrastructure such as ditches and field ridges. In some areas, crop residues are retained to cover the ground surface. This ecosystem not only provides local food and cash crop output, but also fixes carbon sinks through photosynthesis of crops and reduces soil erosion through crop residue mulching

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The proposed project activities are intended to follow all applicable legal and regulatory requirements of local, state and national policies. There are no laws today in China that prevent the implementation of the proposed project activity. The following legislation are considered for the project activity:

The following national laws and regulations have been identified as associated with the project: Constitution of the People's Republic of China⁵, Agriculture Law of the People's Republic of China⁶, Land Management Law of People's Republic of China⁷, Law of the People's Republic of China on Land Contract in Rural Areas⁸, Production Safety Law of the People's Republic of China⁹, Labour Law of the People's Republic of China¹⁰, Regulations on the Protection of Basic Farmland¹¹.

The project also complies with the following provincial regulation and national/industry standards:

- Regulations of Inner Mongolia for the Administration of Farmland Construction Projects¹²
- Land Spatial Planning of Inner Mongolia (2021-2035)¹³
- Regulations of Inner Mongolia on Land Administration¹⁴
- Inner Mongolia rural Industry development "14th Five-Year Plan"¹⁵

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

⁵ https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

⁶ http://www.gov.cn/ziliao/flfg/2005-09/12/content_30998.htm

⁷ http://www.gov.cn/banshi/2005-05/26/content_989.htm

⁸ http://www.zfs.moa.gov.cn/flfg/202002/t20200217_6337175.htm

⁹ https://www.mem.gov.cn/fw/flfgbz/fl/202107/t20210716_392186.shtml

¹⁰ http://www.gov.cn/banshi/2005-05/25/content_905.htm

¹¹ http://www.gov.cn/zhengce/2020-12/26/content_5574284.htm

¹² http://nmt.nmg.gov.cn/gk/zfxxgk/fdzdgknr/zcfg/202111/t20211125_1961999.html

¹³ <https://new.qq.com/rain/a/20210630a0d15o00>

¹⁴ http://www.nmgrd.gov.cn/zyfbx/tzggx/202210/t20221010_408768.html

¹⁵ https://www.nmg.gov.cn/zwgk/zfxxgk/zfxxgkml/ghxx/zxgh/202204/t20220418_2040496.html

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes

☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes

☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes

☒ No

1.18 Sustainable Development Contributions

The Project provides many benefits that will help achieve China's Sustainable Development Goals, a set of 17 universal goals covering the thematic areas of environmental, economic, and social development.

During the implementation of the project, it is estimated to achieve 3 SDGs as follows:



Goal 4 Quality Education: In order to implement improved agricultural land management, farmers were trained and educated, including the principle, technical operation method, monitoring method of improve organic fertilizer and soil conditioners application, improve water management/irrigation, reduce tillage/improve residue management. During each monitoring period, training record will be provided.



Goal 15 Life on land: The project strives to achieve a land degradation-neutral world, while it also has a positive impact on the ecosystem of the project land as the soil disturbance is mitigated. The project activity prevented 37,272.2 ha of cropland to be degraded. During each monitoring period, proportion of land that is degraded over total land area in project district will be provided.



Goal 13 Climate Action: By replacing the traditional farming and large-scale field irrigation models with the application of organic fertilizers and soil conditioners, large-scale straw returning to the field and water-saving irrigation, the project achieves significant GHG reduction and removal, hence contributing to climate change mitigation. During each monitoring period, the GHG reduction and removal will be calculated according to the applied methodology VM0042.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

According to paragraph 8.4 of VM0042 (version 2.1) , improved ALM projects may result in leakage through: new application of organic fertilizers and soil conditioners from outside the project area (i.e., organic amendments applied in the project from outside of the project area, that were not previously applied in the historical look-back period); productivity declines; displacement of livestock outside of the project boundary; and/or diversion of biomass residues that were used for bioenergy applications in the baseline scenario.

The project does not involve manure treatment, productivity declines, the displacement of livestock outside of the project boundary or diversion of biomass residues that were used for bioenergy applications in the baseline scenario. Thus, it is not applicable to this project.

1.19.2 Commercially Sensitive Information

None of the project documents will be considered as commercially sensitive information, and all the documentation is available to any stakeholders.

1.19.3 Further Information

There is no further information that may have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

2.2 Risks to Stakeholders and the Environment

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

2.3 Respect for Human Rights and Equity

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

2.4 Ecosystem Health

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0042	Methodology for Improved Agricultural Land Management	2.1

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0042	Projects must introduce or implement one or more new changes to pre-existing agricultural management practices which: a) Reduce fertilizer (organic or inorganic) application; b) Improve water management/irrigation; c) Reduce tillage/improve residue management; d) Improve crop planting and harvesting (e.g., improved agroforestry, crop rotations, cover crops); and/or e) Improve grazing management. Appendix 1 provides a non-exhaustive list of eligible ALM practices. A change in practice constitutes adoption of a new practice, cessation of a pre-existing practice or adjustment to a pre-existing practice that results in GHG emissions reduction or removal.	The agricultural management practice prior to the project initiation was primarily conventional tillage with broad irrigation. The main categories of project practice include the application of organic fertilizers and soil conditioners, straw returning to the field, and the construction of water-saving irrigation measures, which belong to option a) b) and c) in accordance with Appendix 1.
VM0042	Projects that introduce or implement quantitative adjustments (e.g., decrease in fertilizer application rate) must exceed five percent of the pre-existing value, calculated as the average value over the historical look-back period,	Before the implementation of the project, organic fertilizers had not been applied on a large scale, and chemical fertilizers were mainly applied to cultivated land. This project has

	developed for the baseline schedule of activities (see Section 6). Appendix 1 gives additional details and guidance on practices and on determining practice change.	effectively enhanced soil fertility by scientifically applying organic fertilizers throughout the entire area and applying soil conditioners to severely salinized soil.
VM0042	Project activities must be implemented on land that is either cropland or grassland at the project start date and remains cropland or grassland throughout the project crediting period except under the following scenarios: a) Introduction of temporary grassland into cropland where it is demonstrated, prior to the project start date and to the addition of new project activity instances, that the integration of forage crops (e.g., annual/perennial grasses, legumes) into annual crops is planned as part of a long-term agricultural management system (e.g., Integrated Crop-Livestock System). Project proponents must provide documentation on the long-term management plans that cover the duration of the project, that describe proposed practices, crops and expected benefits and outcomes of integrated grassland-cropland management; or b) A one-time conversion from grassland to cropland or vice versa where it is demonstrated prior to project validation that project lands in the baseline scenario are degraded and the introduction of improved land use change practices would significantly improve soil health. project proponents must provide documentation demonstrating that lands are degraded at the start of the project and degradation will continue in the baseline scenario due to the presence of degradation drivers or pressures in the baseline scenario. See Appendix 2 for procedures on how to propose this type of land use change.	The project is implemented on the cultivated land within the territory of Tumend Left Banner. The project area was cropland at the project start date and will remain cropland throughout the project crediting period; no land use change will occur.

VM0042	Empirical or process-based models used to estimate stock change/emissions via Quantification Approach 1 must be: a) Publicly available, though not necessarily free of charge, from a reputable and recognized source (e.g., the model developer's website, IPCC or government agency). Sufficient conceptual documentation of inputs, outputs and information on how the model functionally represents SOC dynamics must be accessible to the public. Providing the source code or an API for independent replication of calculations is not required; b) Shown in peer-reviewed scientific studies to successfully simulate changes in SOC and trace gas emissions resulting from the changes in ALM practices included in the project description; c) Able to support repetition of the project model simulations. This includes clear versioning of the model used in the project and stable software support, as well as fully reported sources and values for all parameters used with the project version of the model. Where multiple sets of parameter values are used in the project, clearly identify the sources of varying parameter sets and how they were applied to estimate stock change/emissions in the project. Acceptable sources include peer-reviewed literature and statements from appropriate expert groups that demonstrate evidence of expertise with the model via authorship of peer reviewed model publications or authorship of reports for entities	a) The programme initially uses the RothC model for the quantification of increased carbon stock in the soil, attributable to the adopted IALMs. This is a publicly available model and has been demonstrated to be used for exactly this purpose at global level. It is freely available from the Rothamsted Research¹⁶ website. b) The project proponent's choice for using the RothC model as a point of departure for quantification of carbon stock, is based on peer-reviewed scientific studies that demonstrate the model's applicability for this purpose¹⁷. c) The project proponent has developed a model calibration and validation manual which describes: -Sources for parameters datasets used. How parameters are used; -The protocol for using different datasets for calibration and validation. In addition, the manual prescribes that the model shall be calibrated for each zone and validated for each combination of practice category and crop functional group within each zone. In case an alternative model shall be identified to meet the requirements, the above shall too be documented in an updated Model Validation Report. The model is to be validated by an independent institution and
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¹⁶ <https://www.rothamsted.ac.uk/rothamsted-carbon-model-rothc>

¹⁷ RothC is a model for the turnover of organic carbon in non-waterlogged topsoil that allows for the effects of soil type, temperature, soil moisture and plant cover on the turnover process. It uses a monthly time step to calculate total organic carbon ($t\ ha^{-1}$), microbial biomass carbon ($t\ ha^{-1}$) and $\Delta^{14}C$ (from which the equivalent radiocarbon age of the soil can be calculated) on a years to centuries timescale. RothC was originally developed and parameterized from the Rothamsted long-term field experiments - hence the name. It was extended to model turnover in grassland and in woodland and to operate in different soils and under different climates. It has now been widely tested and used at the plot, field, regional and global scales using data from many long-term experiments, different regions, and countries throughout the world. Version 26.3 was used in the validation of this Programme. The Model Validation Report will be provided in the validation process.

	supporting climate-smart agriculture. These sources must describe the datasets and statistical processes used to set parameter values; d) Validated per datasets and procedures detailed in Section 5.2 of VMD0053 Model Calibration and Validation Guidance for the Methodology for Improved Agricultural Land Management. Model prediction error must be calculated using datasets as described in Section 5.2.5 of VMD0053 and must use the same parameters or sets of parameters applied to estimate stock change/emissions in the project; and e) Using the same model version in the baseline and project scenarios. Further, the same parameters/parameter sets must be used in the baseline and project scenarios. Model input data must be derived following guidance in Table 6 and Table 8. Model uncertainty must be quantified following guidance in Section 8.6. Models may be recalibrated or revised based on new data, or a new model may be applied, provided the above requirements are met.	this forms part of the Project Proponent records. d) The model applied will demonstrate how it meets the requirements set out in the VMD0053 “Model Calibration and Validation Guidance for the Methodology for Improved Agricultural Land management”. The model is to be validated by an independent institution and this forms part of the Project Proponent records. e) The same model version is used in the baseline and project scenarios. Further, the same parameters sets are used in the baseline and project scenarios. Model input data is derived following guidance in Table 6 and Table 8 of VM00042(version2.1). Model uncertainty is quantified following guidance in Section 8.6 of VM00042 (version 2.1).
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Methodology ID	Not applicable under the following conditions	Justification of compliance
VM0042	The project area has been cleared of native ecosystems within the 10-year period immediately prior to the project start date.	The project area has not been cleared of native ecosystems (cropland) within the 10-year period prior to the project start date.
VM0042	The project activity is expected to cause a sustained reduction in productivity of greater than 5% ¹⁸ , as demonstrated by peer-reviewed and/or published studies on the activity in the region or a comparable region.	The improved agricultural land management can increase the crop yield based on the recording of TAAHB. Thus, the project does not result in a

¹⁸ 5% is the threshold for emissions that can be considered de minimis (see VCS Methodology Requirements v4.4, Section 3.3.7).

		sustained reduction of greater than 5% in productivity.
VM0042	The project activities occur on a wetland. this condition does not exclude crops subject to artificial flooding where it is demonstrated that crop cultivation does not impact the hydrology of any nearby wetlands.	The project activity occurs on upland corn and sunflower fields and does not occur on any wetland.
VM0042	The project activity is a biochar application. Biochar may be applied as a soil amendment in the project area provided that the total organic carbon content of the biochar applied is subtracted from the estimated SOC stock change in the project scenario at each verification event.	The project does not include application of biochar as a soil amendment.

As described above, this methodology VM0042 (version 2.1) is applicable to the project.

3.3 Project Boundary

According to the applied methodology VM0042 (Version 2.1), the selected carbon pools included in the project boundary in the baseline and project scenarios are listed in Table 3-1 below.

Table 3-1 Selected Carbon Pools in the Baseline and Project Scenario

Source	Included?	Justification/Explanation
Aboveground woody biomass	No	The project activities are mainly aimed at crops and do not involve woody plants, so it is not included.
Aboveground non- woody biomass	No	According to methodology, this carbon pool is not included
Belowground woody biomass	No	The project activities are mainly aimed at crops and do not involve woody plants, so it is not included.
Belowground non- woody biomass	No	According to methodology, this carbon pool is not included
Deadwood	No	According to methodology, this carbon pool is not included

Litter	No	According to methodology, this carbon pool is not included
Soil organic carbon	Yes	Major carbon pool affected by project activity that is expected to increase in the project scenario.
Wood products	No	According to methodology, this carbon pool is not included

According to the applied methodology VM0042 (version 2.1), the spatial extent of the project boundary encompasses all lands subject to implementation of the proposed improved agricultural land management practices.

Therefore, the project boundary includes all the farmlands within the IALM project in Tumed Left Banner. The map of the project boundary is provided in Figure 1-1 of section 1.13; a KML file is also submitted for identification of the project boundary.

Selected GHG Sources in the Baseline and Project Scenario are listed in Table below.

Source	Gas	Included?	Justification/Explanation
Soil organic carbon	CO ₂	Yes	This is the primary source of emission removals that the project is aiming to achieve. Quantified as stock change in the pool, rather than an emissions source.
Fossil fuel	CO ₂	No	This project does not generate additional use of agricultural machinery and is consistent with the baseline scenario.
Soil methanogenesis	CH ₄	No	Excluded from the project, as model used for SOC turnover does not have an exclusive soil methanogenesis element in calculations. It has been excluded from both the baseline and project activity.
Enteric fermentation	CH ₄	No	The project does not involve any livestock.
Manure deposition	CH ₄	No	The project does not use any manure as fertilizer.
	N ₂ O	No	The project does not use any manure as fertilizer.
Use of nitrogen fertilizers	N ₂ O	No	The nitrogen fertilization is lower in the with-project scenario relative to the baseline scenario, N ₂ O emissions from nitrogen fertilizers are not included in the project boundary.
Use of nitrogen fixing species	N ₂ O	No	The project does not involve nitrogen fixing species.

Biomass burning	CO ₂	No	The project does not burn any biomass.
	CH ₄	No	The project does not burn any biomass.
	N ₂ O	No	The project does not burn any biomass.
Woody biomass	CO ₂	No	The project activity does not involve woody biomass.

3.4 Baseline Scenario

Development of schedule of activities in the baseline scenario

According to the applied methodology VM0042 (version 2.1), continuation of pre-project ALM practices is the most plausible baseline scenario. Following the methodology guidance, the baseline scenario is determined by producing an annual schedule of activities from the practices applied in each field/farming unit/project activity instance, from a historic look-back period (of at least three years or to include one complete crop-cycle). Baseline emissions/stock changes are then modeled (Quantification Approach 1) or (for SOC stock change only) directly measured in baseline control sites subject to the annual schedule of activities (Quantification Approach 2). Note that under Quantification Approach 1, direct SOC stock estimates are also required at $t=0$ to serve as model input for model initialization. The emission removals of the project are estimated via Quantification Approach 1.

The geographic area of the project is divided into smaller areas (strata) which are homogeneous with respect to climatic and edaphic conditions. This is to reduce the required number of soil samples to be taken for quantification of baseline emissions without increasing uncertainty. The information on the characteristics of each stratum is obtained from Tumed Left Banner Agricultural and Animal Husbandry Bureau.

The crops grown in this project site are harvested once a year. Annual tillage and fertilization operations are carried out in both the baseline year and the project year, which are regarded as annual cycles. Therefore, the baseline practice assessment covers a three-year historical retrospective period.

3.5 Additionality

The Project uses a project method for the demonstration of additionality in accordance with the applied methodology VM0042 version 2.1 and VCS Methodology Requirements version 4.4. Refer to Section 3.5.1 for Step 1 (Regulatory Surplus).

Refer to Section 3.5.2 for Step 2 and Step 3.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

The project is located inside a Non-Annex 1 country and the project activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

As per the latest version of the VCS Standard, section 3.14.1 and methodology VM0042v2.1 requirements, the additionality of this project is demonstrated in two steps:

1. The project shall demonstrate regulatory surplus at validation and each project crediting period renewal.

A. Regularity surplus

B. Barrier analysis

C. Common practice analysis

2. Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project.

3.6 Methodology Deviations

No deviations from the procedures indicated by the methodology have been made.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

4.2 Project Emissions

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

4.3 Leakage Emissions

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 - Pipeline Listing Process.

5 MONITORING

5.1 Data and Parameters Available at Validation

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 – Pipeline Listing Process.

5.2 Data and Parameters Monitored

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 – Pipeline Listing Process.

5.3 Monitoring Plan

This section has been omitted in accordance with the VERRA Registration and Issuance Process v4.6 October 16, 2024 - § 3.1.3 – Pipeline Listing Process.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

The project is to be listed, this section is not required.